

EXPRESSIONS OF GASEOUS CONCENTRATION AND DILUTION RATIOS

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Abstract—Ratios of dilution or concentration of a gas sample are expressed as a *pollutant quotient*, or, if appropriate, an *odorant quotient*, Z , denoting the ratio of a pollutant (or odorant) concentration to some target concentration. The expression pZ is defined as $+\log_{10}Z$, so that a positive value of pZ implies dilution and a negative value concentration. The effect of particulate matter on the Z function is considered. Various examples are given, including conversions from other common dilution expressions.

INTRODUCTION

THERE are many occasions in the sampling of gases and vapors for sensory evaluation or for chemical analysis that require some change of concentration. For example, an increase of concentration may be needed to reach the detectable range, or a decrease (dilution) to render a sample tolerable for human exposure, or to establish threshold limits. The literature is replete with descriptions of devices and procedures to effect such changes, but there is no uniform mode of expression of concentration ratios, and some of the expressions are, indeed, confusing, especially in the field of odor measurement.

It therefore seems worthwhile to adopt a standard set of expressions for such ratios.

THE POLLUTANT QUOTIENT

The fundamental dimensionless expression is the *pollutant quotient*,*

$$Z = C/C_t,$$

where

C = the concentration of pollutant in a homogeneous gas sample. Concentration may be expressed as mass of pollutant per volume of sample (such as mg m^{-3}) or volume of pollutant per volume of sample (such as ppm by volume), where the volumes of pollutant and of sample are both referred to the same conditions of temperature and pressure.

C_t = the target concentration to be reached by diluting or concentrating the sample. Units must be the same as those used for C . When the target concentration is some functionally defined sensory odor threshold, then C_t denotes the threshold concentration.

When $Z > 1$ the sample must be diluted to reach the target concentration, and Z may be called the *pollutant dilution quotient*, or simply the *dilution quotient*.

* The letter Z is after ZWAARDERAKER (1926) who was the earliest investigator to use dilution ratios for odor measurements. Where the primary pollutant effect is odor, Z may be called the *odorant quotient*.

When $Z < 1$ the sample must be concentrated to reach the target concentration, and Z may be called the *pollutant concentration quotient*, or simply the *concentration quotient*.

We also define the expression (analogous to pH)

$$pZ = +\log_{10} Z.$$

Example 1

The concentration of an odorous pollutant in a gas sample is 100 ppm. The target concentration to be reached by diluting the sample is the sensory threshold at 0.10 ppm. Then,

$$Z = \frac{C}{C_t} = \frac{100}{0.10} = 1.0 \times 10^3,$$

and

$$pZ = +\log_{10} Z = +3.0.$$

Note that when pZ is positive, Z is a dilution quotient.

Example 2

The concentration of gaseous pollutant in an ambient air sample is 0.10 ppm. The target concentration is reached by concentrating the sample to 100 ppm for use in chemical analysis. Then,

$$Z = \frac{C}{C_t} = \frac{0.10}{100} = 1.0 \times 10^{-3},$$

and

$$pZ = +\log_{10} Z = -3.0.$$

Note that when pZ is negative, Z is a concentration quotient. The following paragraphs give several expressions that are equivalent to the pollutant quotient, Z .

(a) *Volumetric dilution.* A given volume of gaseous sample, V , is diluted at constant temperature and pressure by adding pollutant-free gas until the target concentration, C_t , is reached. Let the volume of the diluted sample be V_t ; then

$$C = m/V, \text{ where } m \text{ is the mass of pollutant in the sample and } C_t = m/V_t.$$

By division $C/C_t = V_t/V$.

Therefore $Z = V_t/V$. In this form Z is a dilution quotient. It is assumed that mass is conserved in the dilution, and that the sample remains homogeneous.

(b) *Dilution by change of flow rate.* A gas stream is flowing at a given rate, Q (volume per unit, such as $\text{m}^3 \text{ min}^{-1}$). The rate is increased by adding pollutant-free gas at constant pressure and temperature until the target concentration, C_t , is reached. Let the new flow rate be Q_t . Then,

$$Q = V/t, \text{ where } t \text{ is time, and } Q_t = V_t/t.$$

By division,

$$\frac{Q_t}{Q} = \frac{V_t}{V}.$$

Then, from paragraph (a) above,

$$Z = \frac{Q_t}{Q}.$$

In this form Z is also a dilution quotient. Here too it is assumed that mass and homogeneity are preserved.

(c) *Dilution by removal of sample (depressurization).* A sample in a rigid container (constant V) at constant temperature and at a given pressure, P , is diluted by removing some of the sample until the target concentration, C_t , is reached. Let the pressure of the diluted sample be P_t . Under these conditions, the pressure is proportional to the concentration (from the ideal gas law), and

$$\frac{P}{P_t} = \frac{C}{C_t} = Z.$$

In practical applications, P_t , the pressure at the target concentration, should be 1 atm for presentation to human subjects, as shown in the following examples.

Example 3

Assume that the original concentration of odorant in a rigid container at atmospheric pressure is C_o . The container is now pressurized with pure air to P_1 , the contents allowed to mix, and the container is depressurized back to 1 atm by venting. The concentration is not diminished by pressurization, but only by venting, which actually removes odorant from a fixed volume. Let the final concentration be C_1 . Then

$$P_1/1 \text{ atm} = C_o/C_1 = Z_1$$

and

$$C_1 = C_o/P_1,$$

where P_1 must be expressed in atmospheres (absolute, not gage pressure) and Z_1 is the first odorant dilution quotient. The container may then be repressurized to a new pressure, P_2 , which when vented gives another reduced concentration, C_2 . Then,

$$C_2 = C_1/P_2 = C_o/P_1P_2,$$

and

$$C_o/C_2 = Z_2 = P_1P_2.$$

Similarly, for any number of successive pressurizations $P_1, P_2 \dots P_n$ and ventings, the final diluted concentration, C_n , is

$$C_n = C_o/P_1P_2 \dots P_n,$$

and

$$C_o/C_n = Z_n = P_1P_2 \dots P_n.$$

For the simple case where each pressurization is 14.7 lb in^{-2} (gage), or 2 atm (absolute), the relationship becomes

$$C_n = C_o/2^n.$$

Then

$$Z_n = 2^n \text{ and } pZ_n = +0.301n$$

where n is the number of pressurization and venting cycles.

Example 4. (Experimental)

A 1000 in^3 (16.4 l.) stainless steel tank fitted with two needle valves and a pressure gage was evacuated and then loaded with cumene (unpurified commercial product) by opening one valve to atmosphere and introducing a known quantity of cumene

vapor; the intruding ambient air served as the carrier (FIG. 1). Initial cumene concentration, C_o , was 340 ppm (vol/vol). The container was pressurized to 4.0 atm (absolute), and then vented. Judges who were familiar with the cumene odor smelled the vented air to establish whether or not cumene odor was recognizable. The threshold level was reached at the fourth venting. Then

$$C_t = C_o/4.0^4 = 340 \text{ ppm}/256 = 1.3 \text{ ppm}$$

$$Z = C_o/C_t = 256, \text{ and } pZ = +2.41.$$

The value was checked by a dynamic dilution system that had been previously validated with SF_6 tracer gas (COLLINS, BARTLETT, TURK, EDMONDS and MARK, 1965). The concentration of cumene in the odorant stream was 85 ppm and the recognition threshold was reached when the flow rates were $23\,530 \text{ cm}^3 \text{ min}^{-1}$ of diluent and $270 \text{ cm}^3 \text{ min}^{-1}$ of odorant. Then,

$$Z = Q_d/Q = C_o/C_t = (23\,530 + 270)/270 = 881,$$

and

$$C_t = 85 \text{ ppm}/881 = 0.97 \text{ ppm}.$$

(d) *Volumetric concentration.* A given volume of gaseous sample, V , is concentrated by reducing its total volume until the target concentration, C_p , is reached. The volume of the concentrated sample is V_t . The expression is the same as that given in paragraph (a) above:

$$Z = \frac{Q_t}{Q},$$

where Z is a concentration quotient.

Example 5.

A sample of air (1000 l.) is passed through an adsorbent that retains all of the pollutants. The pollutant is then desorbed into a 500-cm^3 container to give the desired concentrated sample.

$$Z = V_t/V = 0.500/1000 = 5.00 \times 10^{-4}$$

$$pZ = -\log_{10} (5.00 \times 10^{-4}) = -3.3.$$

The negative sign of pZ confirms that Z in this case is a concentration quotient.

When a sample is characterized by measuring its pollutant quotient, Z , it is assumed that the chemical identity of the pollutant is preserved and that the sample remains homogeneous. In applying such measurements to odor problems in air, these assumptions may be considered to be true in any of the following circumstances:

- (a) when a gaseous sample is diluted in air under conditions such that no chemical reactions occur and no inhomogeneities, such as those which may be related to aerosol particles, distort the pollutant concentrations;
- (b) when a gaseous sample is treated so that a representative portion of its pollutant matter is removed, as by non-selective adsorption, or is converted to non-odorous matter, as by oxidation to CO_2 and H_2O ; or
- (c) when pollutant matter is uniformly dispersed into a sample of air, as when a given mass of pollutant evaporates into a closed space.

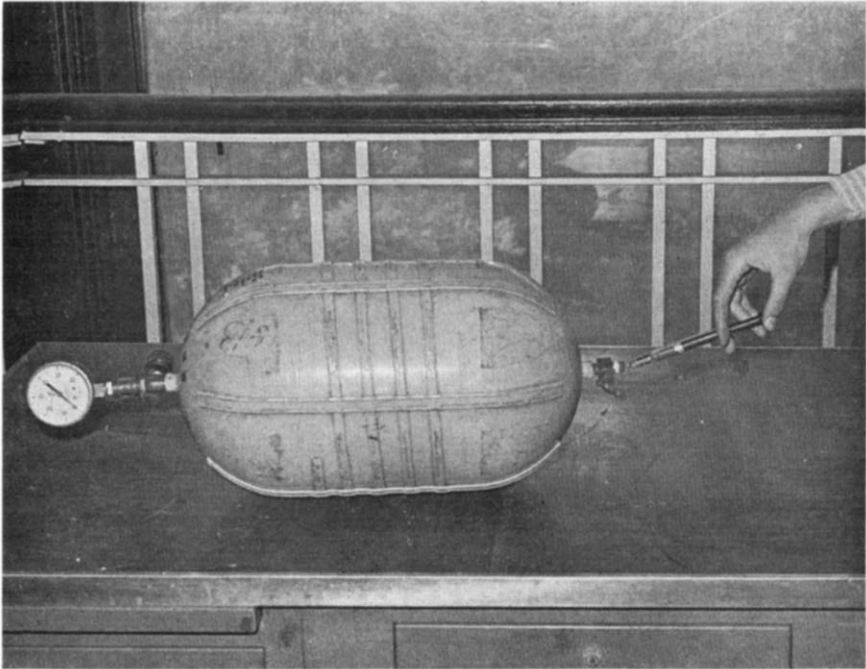


FIG. 1 Loading rigid container with odorant vapour

(Facing p. 970)

CHANGES IN GAS CONCENTRATION IN THE PRESENCE OF PARTICULATE MATTER

Consider a heterogeneous system containing pollutant both in the gas phase and adsorbed on an aerosol. Changes in the concentration of gaseous pollutant will be modified by adsorption or desorption from the aerosol particles. Let the initial conditions be denoted as:

V —volume of sample;

C —concentration of pollutant in the gas phase (mass per unit volume);

C_a —concentration of odorant in the aerosol phase (average mass of pollutant per particle);

C_p —concentration of aerosol (particles per unit volume).

The corresponding target conditions are V_t , C_t , C_{a_t} , and C_{p_t} . Consider the dilution condition. C_a is a positive function of C ; therefore C_a will be greater than C_{a_t} when dilution occurs. Thus, on dilution, pollutant matter will be desorbed from the aerosol particles into the gas phase. Therefore the final (diluted) concentration of pollutant in the gas phase will be the sum of the concentration contributed by the original gaseous matter plus the concentration contributed by pollutant desorbed from the aerosol phase. (the concentration case is just the reverse; the pollutant will be diminished by adsorption). Thus

$$C_t = C(V/V_t) + \frac{\left[\left(\begin{array}{c} \text{mass of odorant adsorbed on} \\ \text{aerosol particles in the sample} \\ \text{volume before dilution} \end{array} \right) - \left(\begin{array}{c} \text{mass of odorant retained} \\ \text{in the adsorbed state} \\ \text{after dilution} \end{array} \right) \right]}{V_t}$$

$$= C(V/V_t) + (C_p C_a V - C_{p_t} C_{a_t} V_t)/V_t.$$

But $C_{p_t} = C_p (V/V_t)$, and substituting this value in the previous equation and simplifying, we have

$$C_t = \frac{V}{V_t} [C + C_p (C_a - C_{a_t})]$$

The equation above implies the following:

- If no aerosol is present, $C_p = 0$ and the equation reduces to the pure gas phase case where $C/C_t = Z = V_t/V$. If an aerosol is present but no adsorption or desorption occurs, $C_a = C_{a_t}$, and again the equation reduces to $Z = V_t/V$.
- If an aerosol is present but desorption is slow ($C_a = C_{a_t}$, momentarily) and the sample is rapidly diluted to volume V_t , the momentary concentration will be $(V/V_t)C$, but this will then increase to

$$\frac{V}{V_t} [C + C_p (C_a - C_{a_t})].$$

Such an increase in odorant concentration may also occur as a result of any other factor that enhances desorption, such as a rise in temperature.

CONVERSION OF OTHER EXPRESSIONS TO THE POLLUTANT QUOTIENT

Typical conversions are illustrated by the following examples.

Example 6

An odorous gas sample is characterized according to the Standard Method for Measurement of Odor in Atmospheres (ASTM, 1967) as having a concentration of 500 odor units ft^{-3} . Since an "odor unit" is defined as "1 ft^3 of air at the odor threshold," the "concentration" is $500 \text{ ft}^3 \text{ ft}^{-3} = V_u/V = Z = 500$.

Thus the ASTM "odor concentration" is converted to "odorant quotient" simply by dropping the words "odor units per cubic feet" to convert the value to a dimensionless number.

Example 7

An odorous effluent from a stack is characterized by the ASTM method as discharging into the atmosphere 1 000 000 odor units per min. The volumetric rate of emission is $1000 \text{ ft}^3 \text{ min}^{-1}$. According to the ASTM standard, the "odor emission rate" equals the volume emission rate times the odor concentration. Then the "odor concentration" of the gas discharged from the stack must be 1000 odor units ft^{-3} and the odorant quotient, Z , is 1000.

Example 8

A sample of water exhibits an "Odor Intensity Index" of 6 (ASTM, 1970). The Index is defined as "the number of times the concentration of the original sample is halved by addition of odor-free water to obtain the least definitely perceptible odor". The odorant concentration in the head gas in equilibrium with the aqueous phase is not a precisely linear function of its mass concentration in the solution. It may be approximately linear in dilute solutions, where mass concentration is roughly proportional to mole fraction, and where the activity coefficient is approximately constant. Under these conditions, each dilution reduces the vapor concentration by a factor of 2, and

$$C/C_i = Z = 2^6$$

and

$$pZ = +\log 2^6 = +1.8.$$

The general conversion formula is

$$\begin{aligned} pZ &= +\log 2^n = +n \log 2 \\ &= +0.301 n, \end{aligned}$$

where n is the ASTM Odor Intensity Index.

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